

AI-Powered Data Cleansing & Diagnostics Report

Goal: analysis of the people age and their alcohol consumption

1. Executive Summary

This diagnostics report details the cleaning operations performed on dataset `alzheimers_disease_data.csv`. Guided by user goals and Nvidia GLM-5.2 recommendations, duplicate/redundant structures were dropped, datatypes were normalized, and missing values imputed.

Dimension	Original Dataset	Cleaned Dataset
Total Rows	2149	2149
Total Features / Columns	35	2

2. Applied Feature Rules

Column	Action	Explanation & Custom Rules
ADL	DROP	Activities of Daily Living (ADL) scores are not relevant to the specific goal of analyzing age and alcohol consumption.
Age	KEEP	This is a primary feature directly required for the user's goal of analyzing age against alcohol consumption.
AlcoholConsumption	KEEP	This is the primary target/feature variable directly required for the user's goal.
BMI	DROP	Body Mass Index is outside the scope of the specific goal focusing on age and alcohol consumption.
BehavioralProblems	DROP	Behavioral problems are not relevant to the specific goal of analyzing age and alcohol consumption.
CardiovascularDisease	DROP	Cardiovascular disease status is not relevant to the specific goal of analyzing age and alcohol consumption.
CholesterolHDL	DROP	HDL cholesterol is outside the scope of the specific goal focusing on age and alcohol consumption.
CholesterolLDL	DROP	LDL cholesterol is outside the scope of the specific goal focusing on age and alcohol consumption.
CholesterolTotal	DROP	Total cholesterol is outside the scope of the specific goal focusing on age and alcohol consumption.
CholesterolTriglycerides	DROP	Triglycerides are outside the scope of the specific goal focusing on age and alcohol consumption.
Confusion	DROP	Confusion symptoms are not relevant to the specific goal of analyzing age and alcohol consumption.
Depression	DROP	Depression status is not relevant to the specific goal of analyzing age and alcohol consumption.
Diabetes	DROP	Diabetes status is not relevant to the specific goal of analyzing age and alcohol consumption.
Diagnosis	DROP	The medical diagnosis (likely Alzheimer's) is not relevant to the specific goal of analyzing age and alcohol consumption.
DiastolicBP	DROP	Diastolic blood pressure is outside the scope of the specific goal focusing on age and alcohol consumption.
DietQuality	DROP	Diet quality is outside the scope of the specific goal focusing on age and alcohol consumption.
DifficultyCompletingTasks	DROP	Difficulty completing tasks is not relevant to the specific goal of analyzing age and alcohol consumption.
Disorientation	DROP	Disorientation symptoms are not relevant to the specific goal of analyzing age and alcohol consumption.
DoctorInCharge	DROP	This column contains anonymized, constant string values ('XXXConfid') providing no analytical value.
EducationLevel	DROP	Education level is not relevant to the specific goal of analyzing age and alcohol consumption.
Ethnicity	DROP	Ethnicity is not relevant to the specific goal of analyzing age and alcohol consumption.
FamilyHistoryAlzheimers	DROP	Family history of Alzheimer's is not relevant to the specific goal of analyzing age and alcohol consumption.
Forgetfulness	DROP	Forgetfulness is not relevant to the specific goal of analyzing age and alcohol consumption.
FunctionalAssessment	DROP	Functional assessment is not relevant to the specific goal of analyzing age and alcohol consumption.

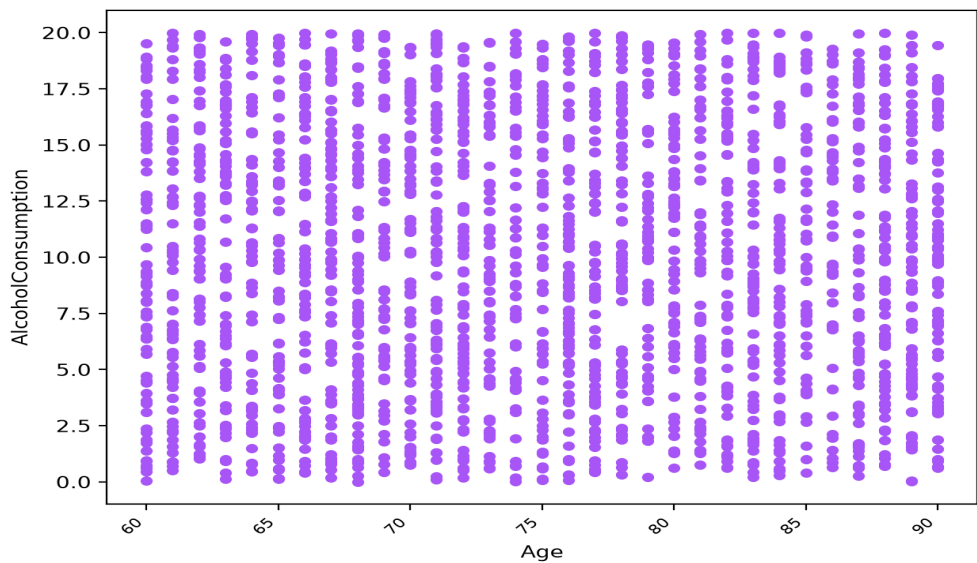
Gender	DROP	While gender can influence alcohol consumption, the user's specific goal focuses strictly on age. It can be dropped to narrow the scope, though it could be kept if the user decides to expand the analysis.
HeadInjury	DROP	Head injury history is not relevant to the specific goal of analyzing age and alcohol consumption.
Hypertension	DROP	Hypertension status is not relevant to the specific goal of analyzing age and alcohol consumption.
MMSE	DROP	Cognitive assessment scores are not relevant to the specific goal of analyzing age and alcohol consumption.
MemoryComplaints	DROP	Memory complaints are not relevant to the specific goal of analyzing age and alcohol consumption.
PatientID	DROP	This is a unique identifier for each patient and does not provide any analytical value regarding age or alcohol consumption.
PersonalityChanges	DROP	Personality changes are not relevant to the specific goal of analyzing age and alcohol consumption.
PhysicalActivity	DROP	Physical activity is outside the scope of the specific goal focusing on age and alcohol consumption.
SleepQuality	DROP	Sleep quality is outside the scope of the specific goal focusing on age and alcohol consumption.
Smoking	DROP	Smoking status is outside the scope of the specific goal focusing on age and alcohol consumption.
SystolicBP	DROP	Systolic blood pressure is outside the scope of the specific goal focusing on age and alcohol consumption.

3. Cleaned Dataset Descriptive Statistics

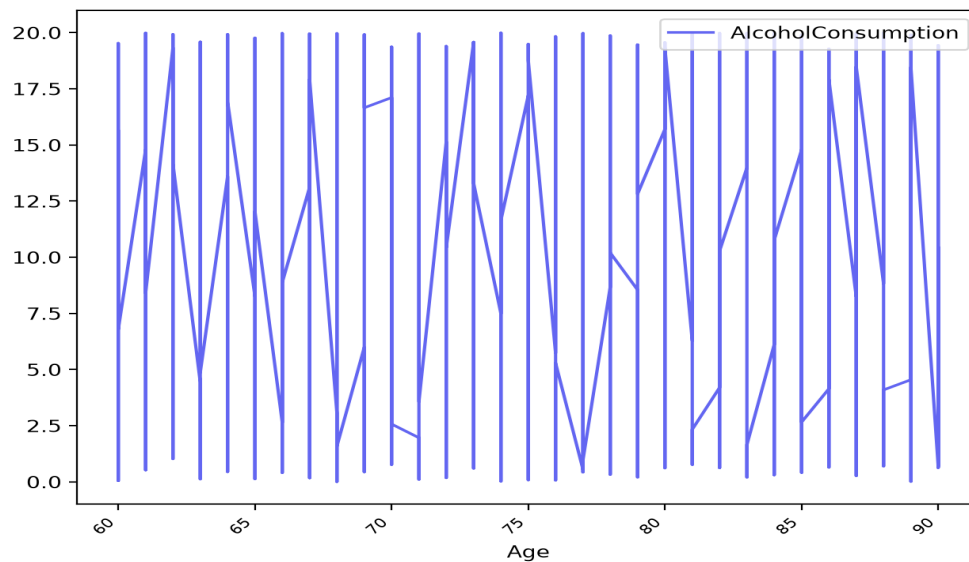
index	Age	AlcoholConsumption
count	2149.0	2149.0
mean	74.91	10.04
std	8.99	5.76
min	60.0	0.0
25%	67.0	5.14
50%	75.0	9.93
75%	83.0	15.16
max	90.0	19.99

4. Diagnostic Visualizations

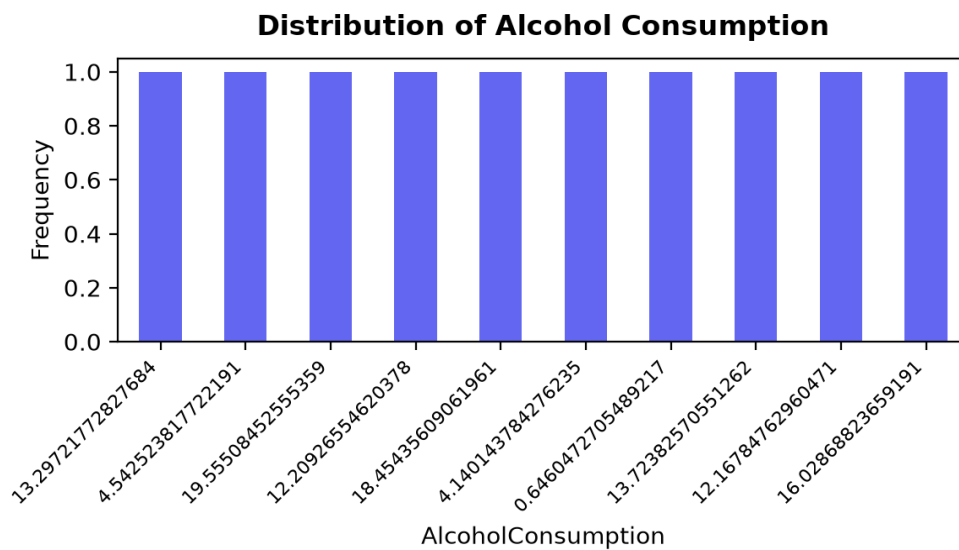
A scatter plot is ideal for visualizing the raw relationship between two continuous variables. It allows us to see the density of data points across different ages and identify any broad trends, clusters, or outliers in alcohol consumption as age increases.



A line plot effectively aggregates the continuous data to show the central tendency (mean) of alcohol consumption for each age. This smooths out individual variances and clearly highlights the overarching trend and trajectory of drinking habits throughout the lifespan.



A histogram provides a clear view of the overall frequency distribution of alcohol consumption values across the entire population. It helps identify the most common levels of consumption, the spread of the data, and whether the distribution is skewed.



A bar plot provides a segmented, easy-to-read comparison of average alcohol consumption across discrete age values. It makes it simple to compare specific age years against one another to pinpoint which ages have the highest and lowest average intake.

Average Alcohol Consumption by Age Group

